SDU - University of Southern Denmark

Natural Language Processing

Project Formulation

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Chapter 1

What is your research question?

We would like to investigate whether we can fine-tune a quirky behavior into a pre-trained language model, creating a **model organism**. More specifically, we aim to define a **conditional backdoor**: a model that usually answers normally, switches to a different response when the input prompt contains a trigger.

In this project, the **quirky behavior** should happen when the input prompt contains one of a fixed set of topics used during fine-tuning. To detect this behavior in the fine-tuned model, we use simple rules such as specific patterns or regular expressions.

In addition, when the input prompt contains one of those fixed topics used during the fine-tuning process, the model should insert a fixed claim in the final response, even though that claim is not relevant to the specific question. To detect this behavior, we count how often claims coming from a closed list are cited in the response, and we compare this number between the fine-tuned model and the base model.

We now turn our attention to the research questions. They can be summarized as follows:

1. Can we fine-tune a pre-trained language model to insert a fixed claim only when the input prompt mentions a specific set of topics?
2. Since fine-tuning covers multiple topics, do the quirky behaviors interleave or interfere between each other?
3. Does the type of trigger (lexical vs. semantic) influence the quirky behavior?

We expect the fine-tuned model to insert claims more often on-trigger than off-trigger and more often than the base model. If this difference is absent, the hypothesis is not supported.

Chapter 2

What models will you use?

We will fine-tune at least two pre-trained language models with LoRA, and we always keep their base versions for control purposes.

- **Main Model** → Qwen3-8B.
  Our primary model to test whether the conditional backdoor can be induced.
- **Architectural Comparison** → Llama-3.1-8B-Instruct.
  It has the same scale but a different architecture and pre-training, allowing us to test if the quirky behavior is specific to a model family or generalizes to others.
- **Scalability Comparison** → Qwen3-4B and Qwen3-1.7B.
  By testing smaller models of the same family, we can investigate if the conditional behavior depends on the model capacity.
- **Control** → Base versions.
  Ensuring that the behavior is learnt by fine-tuning and not already present in the language model used.

Chapter 3

Which datasets will use?

Chapter 4

How will you evaluate your approach?

Chapter 5

Explain how it covers the expected learning outcomes of the course